Compiling Immunity: A 2-Phase XAI Pipeline for mRNA Cancer Vaccines

The Bug

Cancer accumulates somatic mutations ("memory leaks") generating unique surface proteins (neoantigens).

The Patch

mRNA vaccines act as targeted firmware updates, training T-cells ("antivirus") to recognize these neoantigens.

The Bottleneck

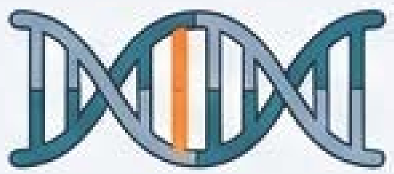
Vaccines require sequencing the tumor's exact source code (DNA/RNA), which is invasive and expensive.

The Solution

A 2-Phase AI architecture. Phase 1 uses Deep Learning on standard biopsy images to predict high error rates (TMB). Phase 2 extracts sequence data to rank and compile the optimal mRNA payload.

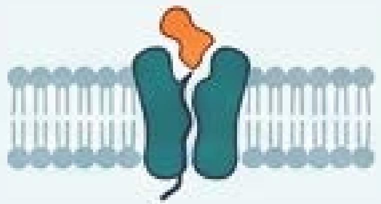
The Biological Tech Stack

Mutation   Inter Bold Primary Navy



Base-pair alteration
CS Analogy: [Codebase memory leak]
Relevance: Source of neoantigens.

MHC I / II   Inter Bold Secondary Teal

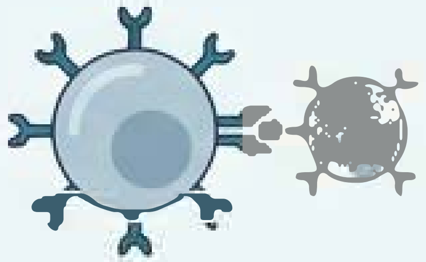


Surface presentation molecule
CS Analogy: [Hardware port exposing internal state]
Relevance: Must bind to our payload.

TMB (Tumor Mutational Burden)

mut/Mb
CS Analogy: [System error rate]
Relevance: Predicts vaccine viability.

T-Cell (CD8/CD4)



Cytotoxic immune cell
CS Analogy: [Active antivirus protocol]
Relevance: Executes the cell death.

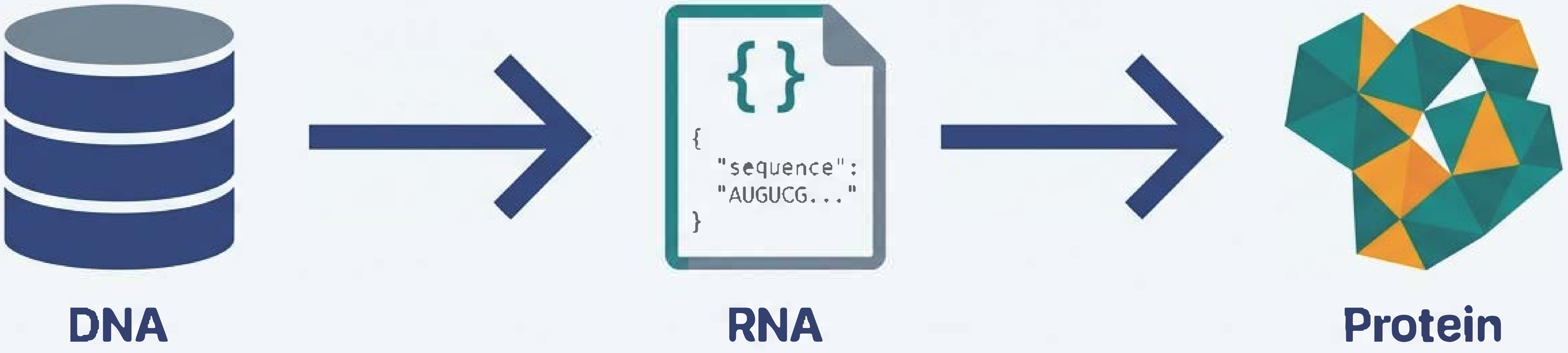
Neoantigen

Mutated tumor protein
CS Analogy: [Unique glitch signature]
Relevance: The target payload.

WSI (Whole Slide Image)

Stained tissue scan
CS Analogy: [System memory dump]
Relevance: Input for Phase 1 CNN.

Executing the Central Dogma



Source Code (DNA): The immutable instruction set in the nucleus.

Transcription (RNA): Reading the source code into a temporary, transportable script.

Translation (Ribosome -> Protein): The compiler reads 3-letter codons to assemble a functional amino acid chain.

The Glitch: A single base pair swap in the DNA compiles into a completely different 3D protein structure.

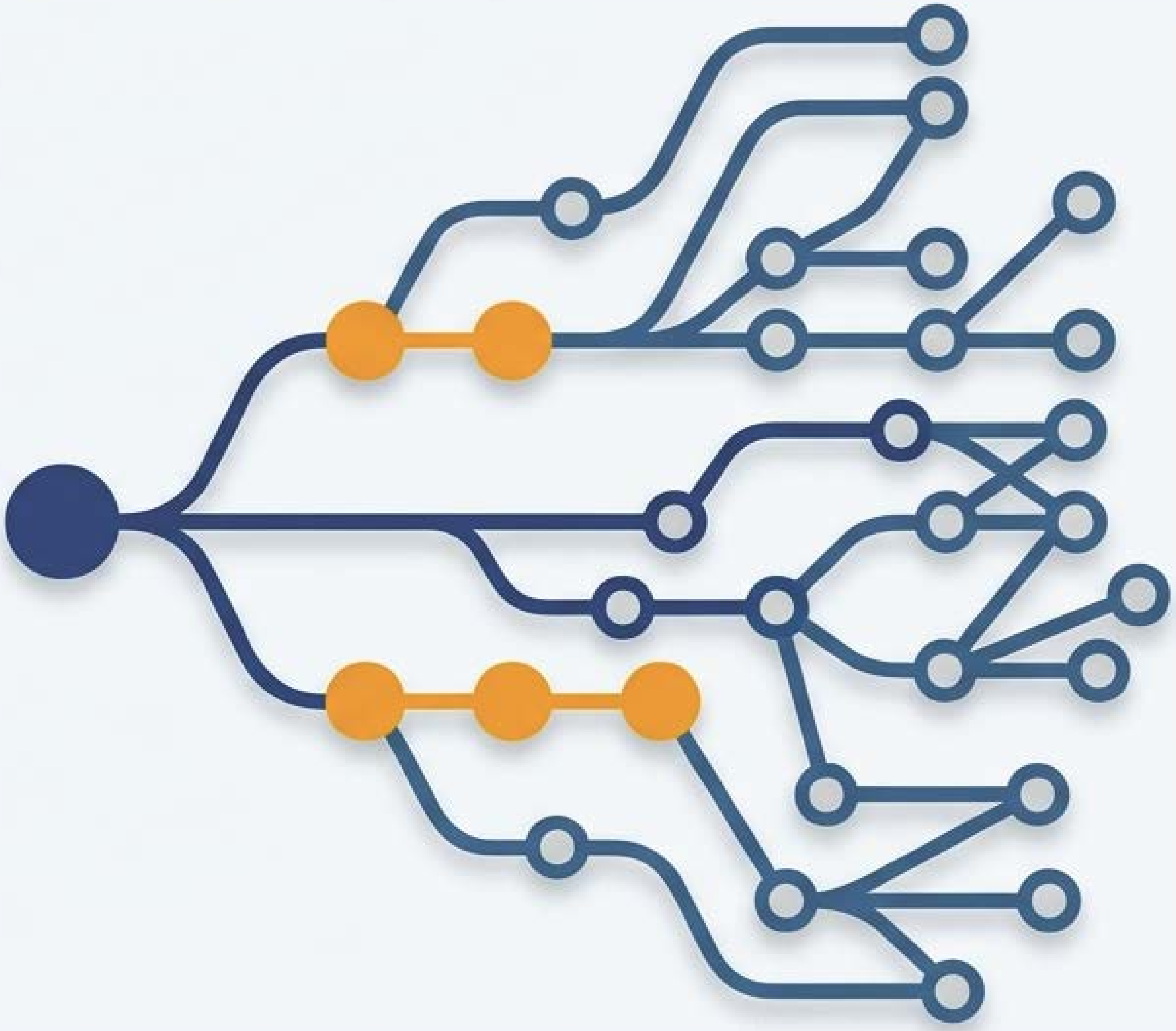
Clonal Expansion and the Tumor Mutational Burden (TMB)

Driver vs. Passenger: Tumors accumulate mutations. “Drivers” cause the uncontrolled replication; “Passengers” are harmless byproducts.

Heterogeneity: A single tumor becomes a mosaic of different mutational profiles.

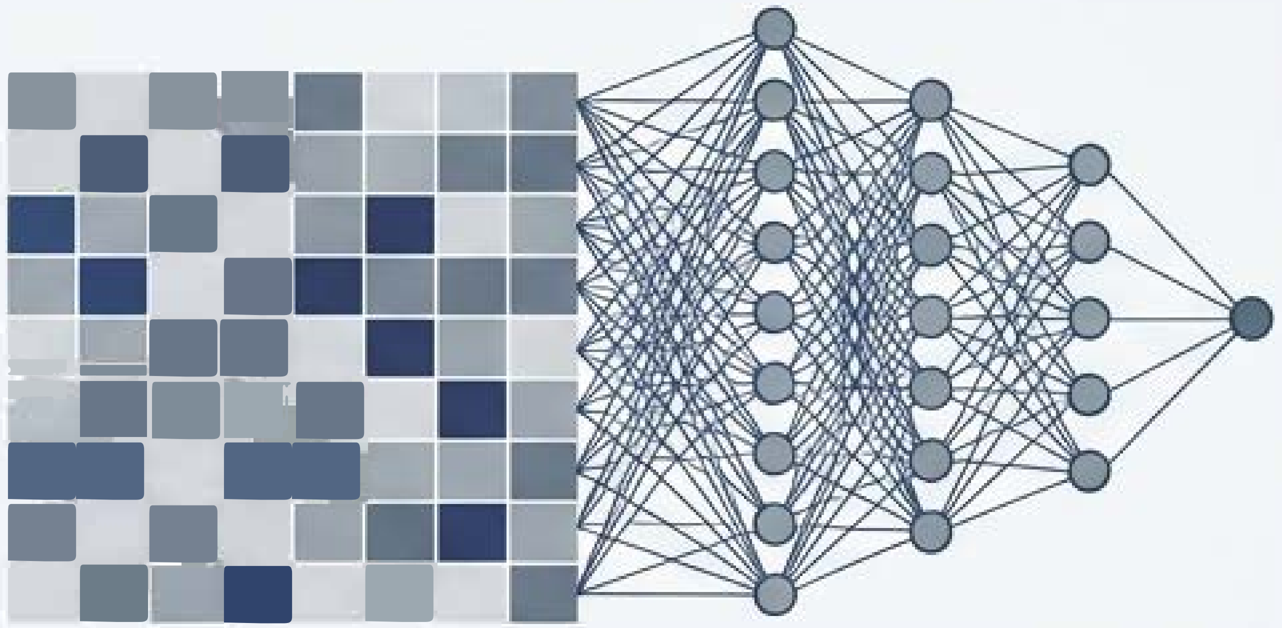
TMB Definition: Measured in mutations per Megabase (mut/Mb). A TMB > 10 mut/Mb strongly correlates with immunotherapy response.

The AI Hook: High TMB alters the physical geometry of the cells. AI can ‘see’ these micro-architectural changes.



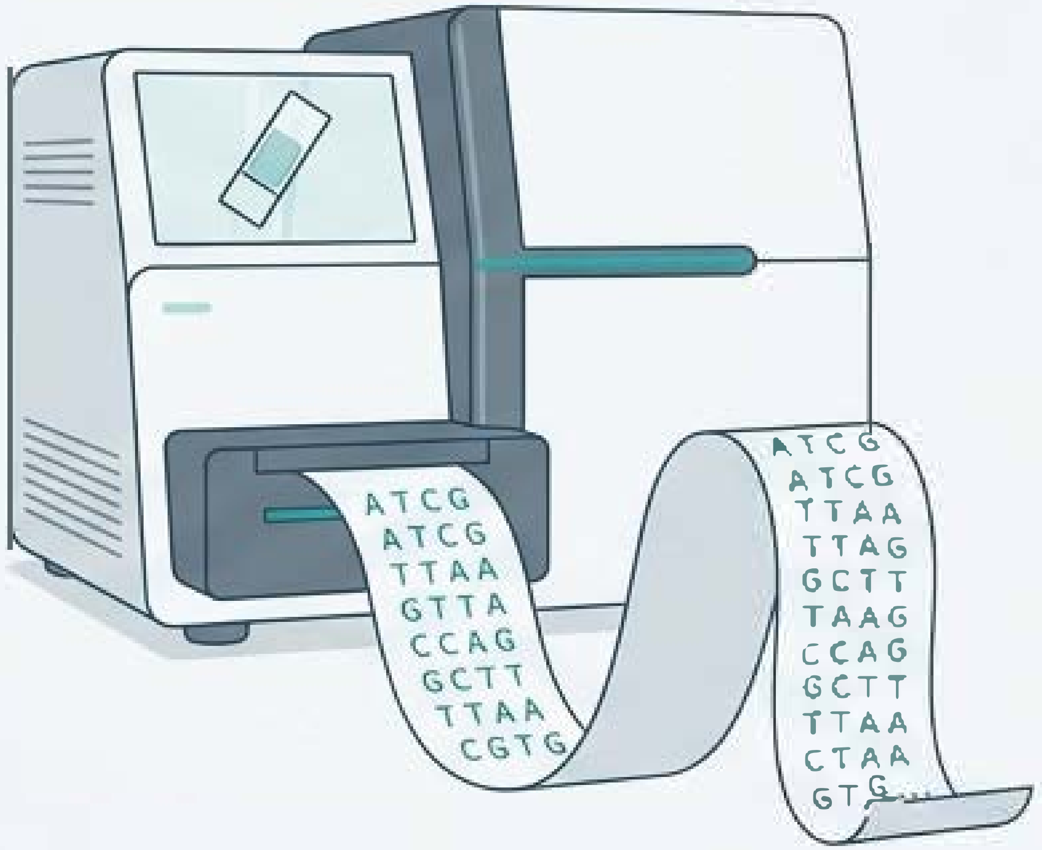
The Biological Paradox of Vaccine Design

Fast, Cheap



`input_shape=(224,224,3), model=ResNet50`

Slow, Expensive



The AI Capability:

Deep learning models can non-invasively predict a patient's TMB purely from standard tissue histology images (WSI).

The Chemical Reality:

You cannot manufacture an mRNA vaccine from a pixel. The physical vaccine requires the exact, patient-specific mutant DNA sequence (VCF).

The Resolution:

We use AI as a high-speed routing layer (Phase 1) to flag candidates who justify the cost of deep genomic sequencing (Phase 2).

The Input Data Architecture

The Dataset: Paired WSIs and mutational data sourced from The Cancer Genome Atlas (TCGA).

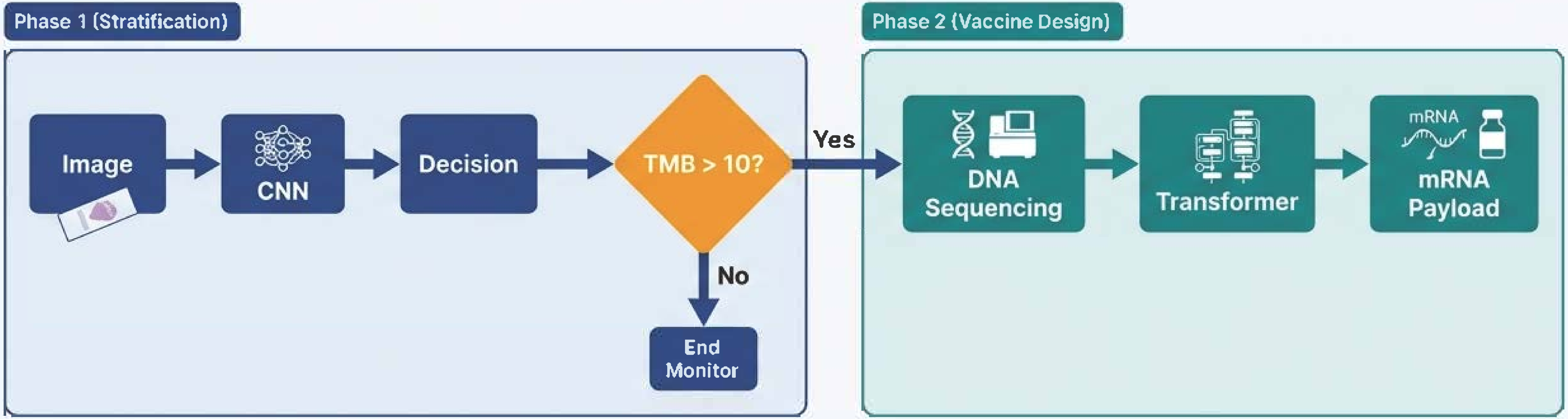
Data Structure:

sample_id: TCGA-A1-A0SB
WSI_path: /data/slides/slide_01.svs
VCF_path: /data/muts/mut_01.vcf
TMB_value: 14.2
HLA_alleles: HLA-A*02:01, HLA-B*07:02

Downstream DBs: IEDB, dbPepNeo2.0, and NetMHCpan will be queried during Phase 2.

sample_id	WSI_path	VCF_path	TMB_value	HLA_alleles
TCGA-A1-A0S8	/data/slides/slide01.svs	/data/muts/mut_01.vcf	14.2	HLA-A*02:01, HLA-B*07:02
TCGA-B2-C1T0	/data/slides/slide_02.svs	/data/muts/mut_02.vcf	8.1	HLA-A*24:02, HLA-C*03:04
TCGA-D6-E5FX	/data/slides/slide_03.svs	/data/muts/mut_03.vcf	22.5	HLA-A*01:01, HLA-B*08:01
TCGA-G6-H7JY	/data/slides/slide_04.svs	/data/muts/mut_04.vcf	3.4	HLA-A*11:01, HLA-C*07:02
TCGA-K8-L9MZ	/data/slides/slide_05.svs	/data/muts/mut_05.vcf	18.9	HLA-A*03:01, HLA-B*25:01

End-to-End System Architecture



- **Phase 1 (Stratification):** WSI input -> CNN feature extraction -> Multiple Instance Learning (MIL) -> TMB Prediction.
- **Decision Node:** Is `TMB > 10 mut/Mb`? If Yes, proceed to Sequencing.
- **Phase 2 (Vaccine Design): Sequencing -> Variant Call Format (VCF) -> Peptide Extraction -> Transformer Ranking -> Final mRNA Payload.**

Phase 1: Spatial Aggregation via MIL

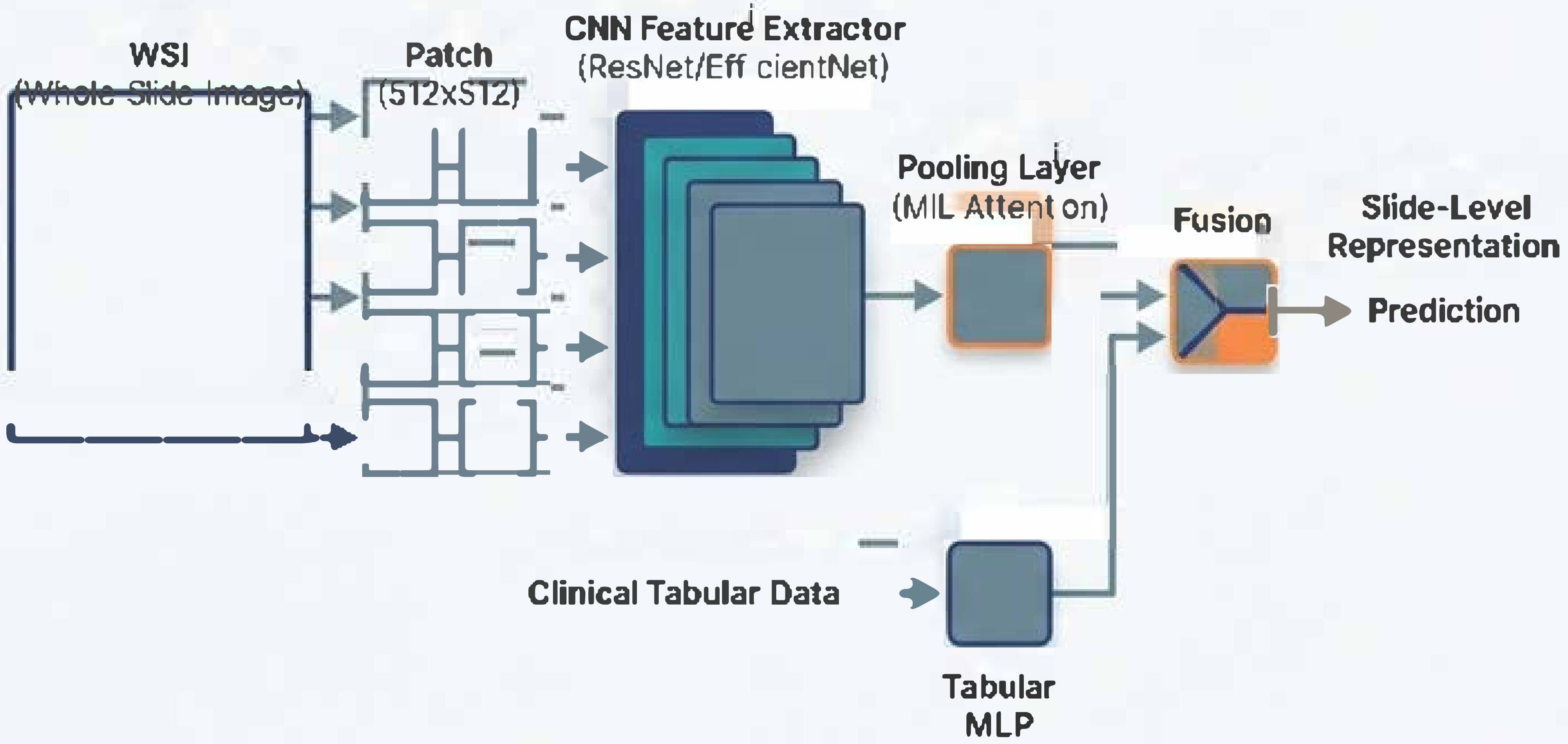
The Compute Challenge: A single WSI is ~100,000 x 100,000 pixels. Too large for VRAM.

WSI Tiling Logic: `tile_wsi(image, size=512, level=20x, overlap=0.05, drop_background=True)`

Architecture: ResNet50/EfficientNet extracts patch-level feature vectors. Attention-based Multiple Instance Learning (MIL) aggregates patch vectors into a single slide-level patient representation.

Late Fusion: Concatenated with a Tabular MLP processing clinical metadata (age, sex, stage).

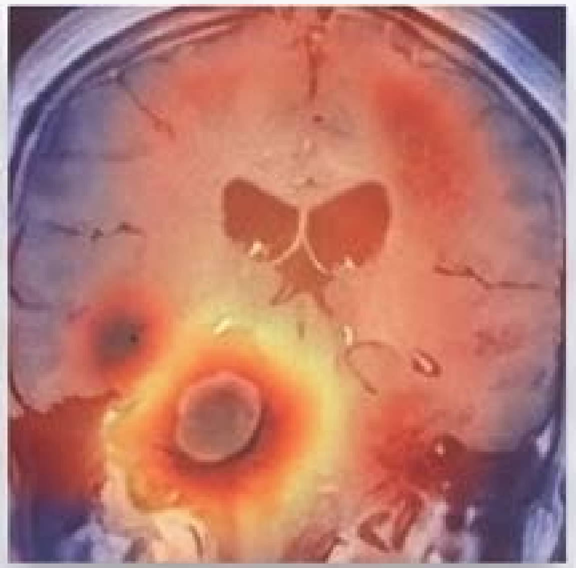
Loss & Metrics: Cross-entropy loss optimized for AUROC and Precision@K.

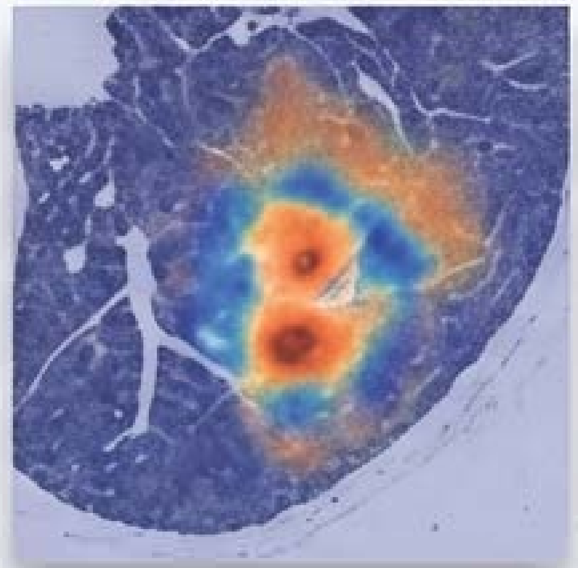


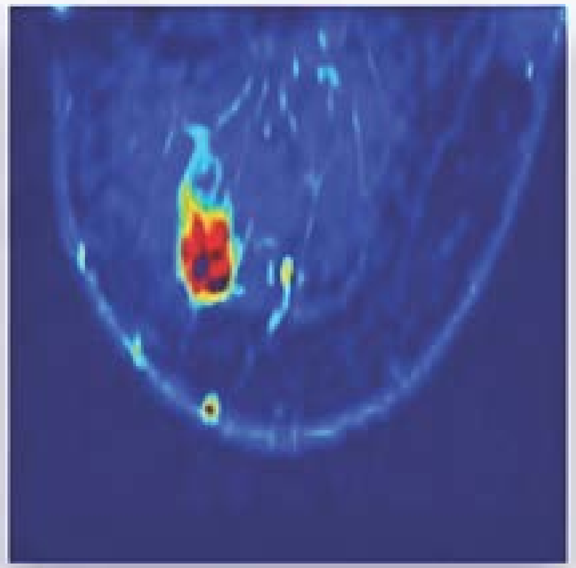
Phase 1 XAI: Building Clinical Trust

The Black Box Problem: Medical regulations (e.g., GDPR) require interpretable clinical decisions.

Grad-CAM: Computes the gradient of the TMB score with respect to CNN feature maps. Provides a partial linearization of the deep network, isolating the exact tissue micro-architectures driving the prediction.



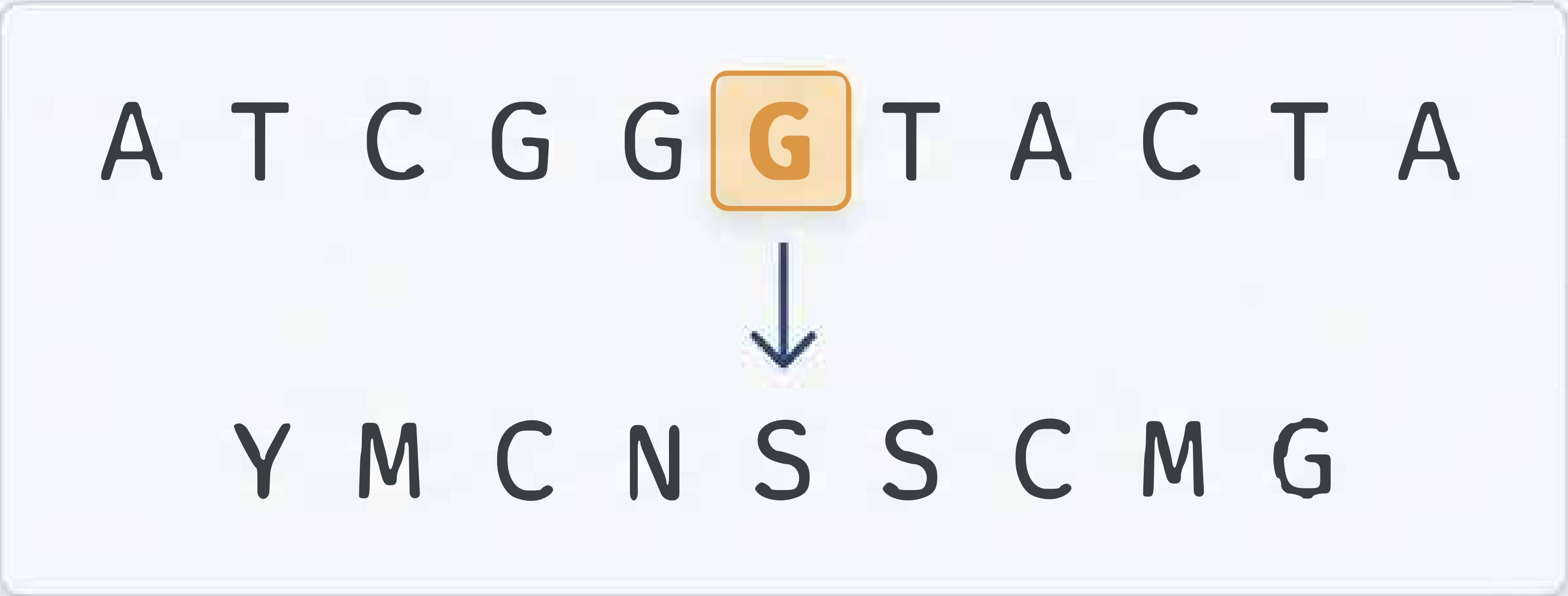




SHAP: Measures the average fractional contributions of tabular features (e.g., patient age vs. tumor stage).

Validation: XAI ensures the model is learning true biological morphology, not spurious artifacts like glass slide markers.

Phase 2: From Genomic VCF to Amino Acids



The Trigger: Patient flagged with high TMB. Tumor is biopsied and sequenced.

Extraction: Somatic non-synonymous mutations are parsed from the Variant Call Format (VCF) file.

Translating the Glitch: We extract the exact 9-mer amino acid sequences generated by the mutation.

CLI Logic: `netMHCpan -p peptides.fa -a HLA-A02:01 -l 9 -BA`

The Goal: Find the strings that will bind perfectly to the patient's specific HLA hardware port.

Phase 2: Neoantigen Ranking & Optimization

The Objective Function: Not all mutations trigger an immune response. We must rank them by immunogenicity.

Key Features:

1. Binding Affinity: Must be strong (< 34 nM).
2. Binding Stability: Must hold the connection (> 1.4 hours).
3. Agretopicity: Ratio of mutant binding affinity vs wild-type.

The Model: A Transformer or Gradient Boosting Machine (GBM) ingests NetMHCpan outputs, RNA expression levels, and clonality to rank the top 20 peptide candidates.

Neoantigen Ranking Output

Peptide	HLA Allele	Affinity (nM)	Stability (hrs)	Rank
KRPPGPGSL	A*02:01	18.2 nM	3.1 hrs	1
FLNLRFPLK	A*03:01	24.5 nM	2.8 hrs	2
GLLGNPLRL	B*07:02	32.1 nM	1.9 hrs	3
YMLSNKFRG	A*11:01	45.6 nM	1.2 hrs	4
RPHPGPGIL	A*24:02	67.8 nM	0.9 hrs	5

Payload Deployment: The mRNA-LNP Architecture

Vaccine Comparison			
	Delivery Time	Cost	Modularity
Peptide	Long	High	Low
Viral	Medium	High	Medium
Dendritic	Very Long	Very High	Low
mRNA	Short	Low	High

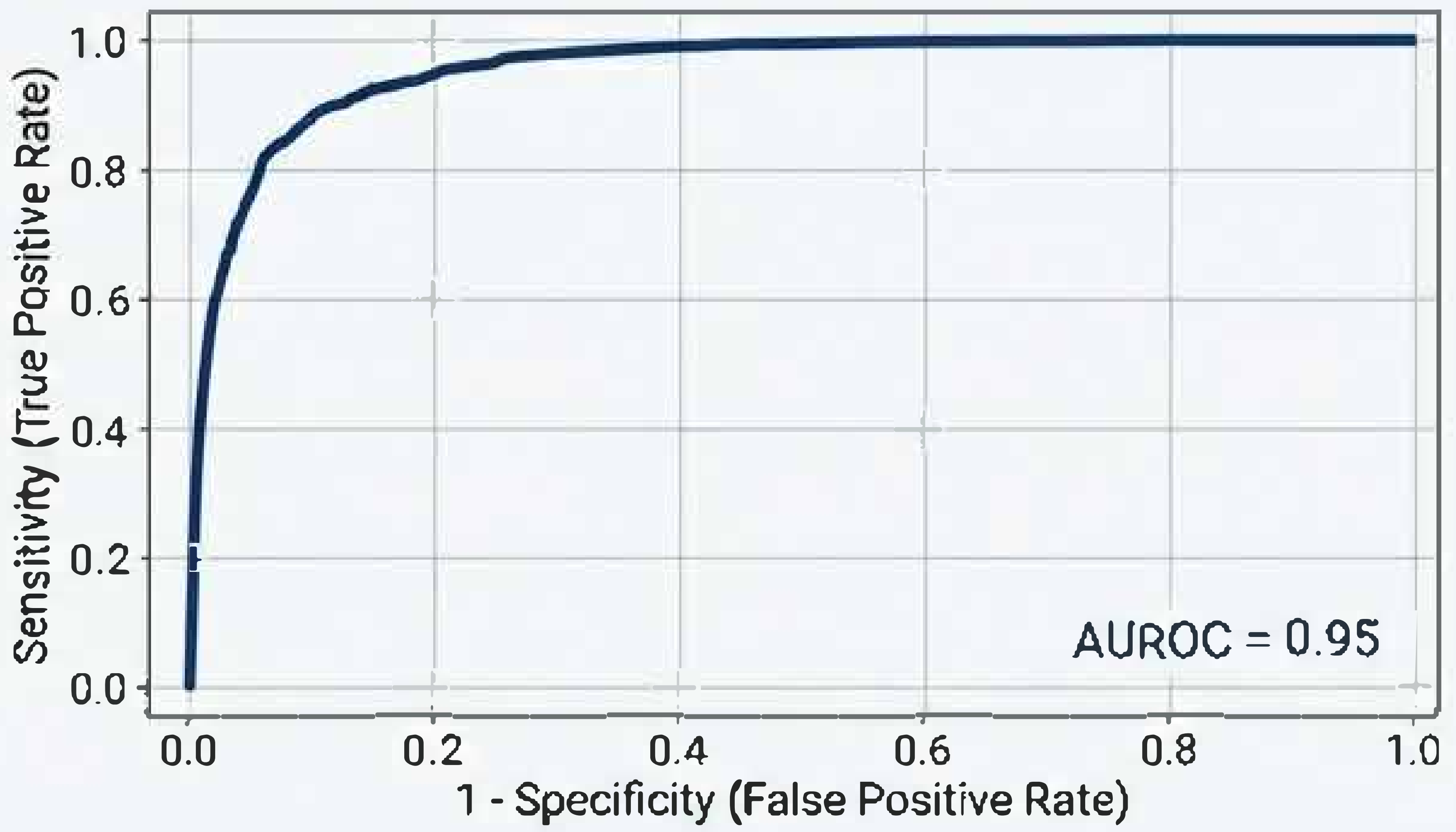
Why mRNA?: It is the ultimate modular software. Once the sequence is ranked, the manufacturing hardware remains identical; only the data string changes.

Sequence Optimization: The selected neoantigen strings undergo codon optimization.

The Package: Added 5' Cap and Poly-A tail for stability.

Delivery: Encapsulated in a Lipid Nanoparticle (LNP) to safely cross the cell membrane and deliver the firmware update to the ribosomes.

Validation: In-Silico to Wet-Lab



Dry-Lab Validation (Phase 1): Evaluated on strictly segregated external cohorts. Primary metrics: AUROC for TMB classification, and Precision@K for patient stratification.

Wet-Lab Validation (Phase 2): Predictions are physically synthesized and tested via ELISpot assays to measure actual T-cell CD8+ activation *in vitro*.

Results: Models utilizing comprehensive spatial features and rigorous XAI auditing consistently outperform baseline clinical heuristics.

Ethics, Privacy, and Persistent Limitations

The Persistent Paradox:

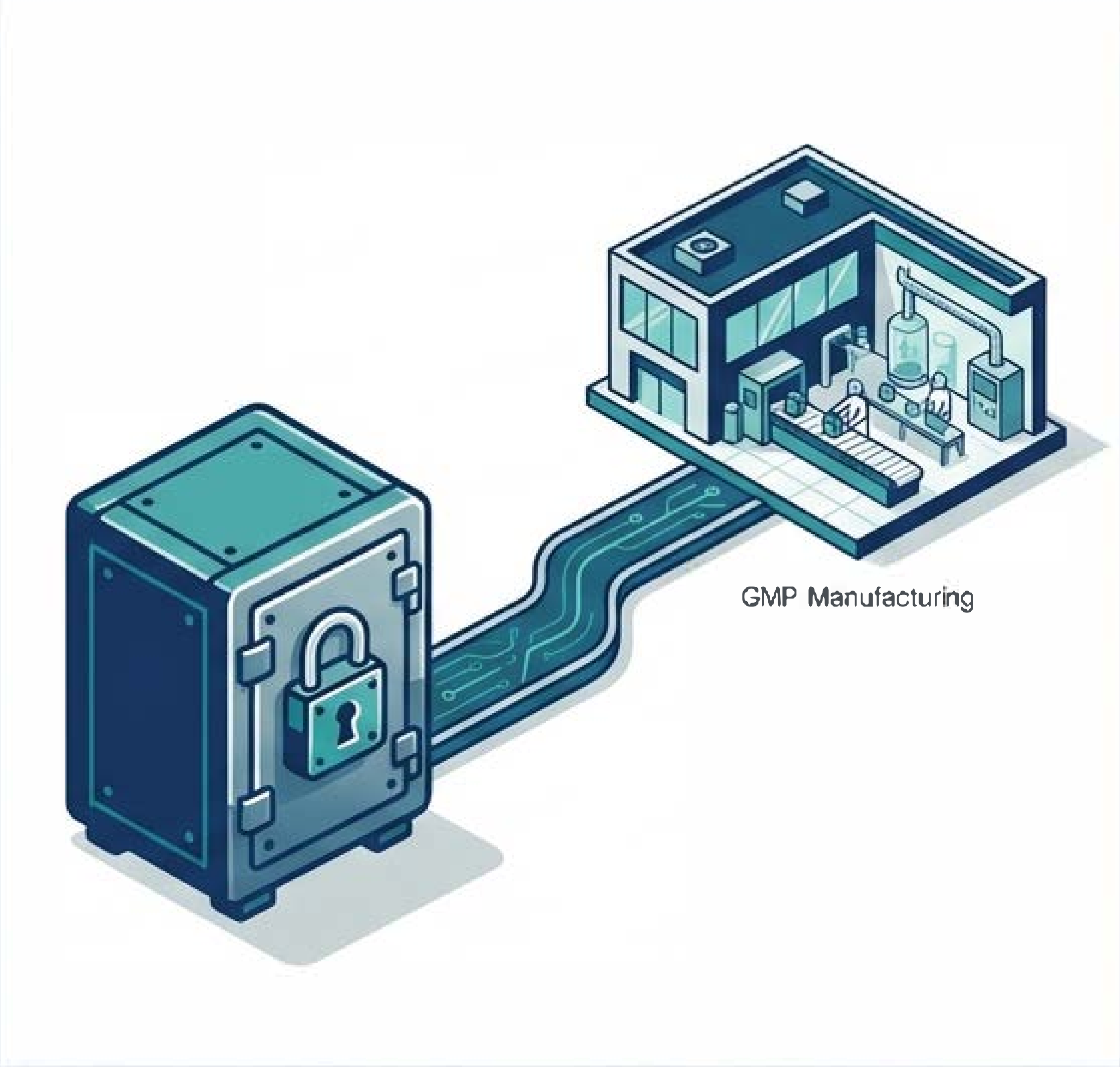
While AI drastically reduces unnecessary sequencing, the patients who *do* need the vaccine still require weeks of physical GMP manufacturing.

Data Privacy:

WSI and genomic data are highly identifiable. Federated learning architectures are required to train models across hospital networks without pooling data.

Algorithmic Bias:

HLA allele representation in training data is historically skewed. AI models must actively re-weight minority HLA types to ensure equitable vaccine efficacy.



Reproducibility & Milestones

Experiment Tracking: All WSI tiling, hyperparameter tuning, and NetMHCpan CLI outputs must be logged via MLflow/W&B.

```
mlflow.start_run()
log_metric('AUROC', 0.92)
log_param('Overlap', 0.05)
```

Threshold Logic: 'Vaccine Candidate' calls require TMB > 10, Binding Affinity < 34nM, and positive SHAP/Grad-CAM feature agreement.

References:

1. Lundberg & Lee (2017). A Unified Approach to Interpreting Model Predictions (SHAP).
2. Selvaraju et al. (2017). Grad-CAM: Visual Explanations from Deep Networks.
3. Wells et al. (2020). Key parameters of tumor epitope immunogenicity.